

Too few, too many, or just right? Optimizing sample sizes for population-level inferences in animal tracking projects

S4. Leave-one out approach

Home range estimation



Figure S4.1. Relative error (%) in home range area estimates across different sampling durations for simulated African buffalos (*Syncerus caffer*) for a leave-one-out approach. Points represent the mean relative error (%) for one of 50 iterations (excluding a different individual each time), with its 95% confidence intervals. Horizontal dotted lines indicate the $\pm 5\%$ error threshold. Blue points fall within this threshold; red points fall outside of it.



Figure S4.2. Relative error (%) in home range area estimates across different sampling durations for simulated Mongolian gazelles (*Procapra gutturosa*) for a leave-one-out approach. Points represent the mean relative error (%) for one of 50 iterations (excluding a different individual each time), with its 95% confidence intervals. Horizontal dotted lines indicate the $\pm 5\%$ error threshold. Blue points fall within this threshold; red points fall outside of it.

Speed & distance estimation



Figure S4.3. Relative error (%) in speed & distance estimates across different sampling intervals for simulated African buffalos (*Syncerus caffer*) for a leave-one-out approach. Points represent the mean relative error (%) for one of 50 iterations (excluding a different individual each time), with its 95% confidence intervals. Horizontal dotted lines indicate the $\pm 5\%$ error threshold. Blue points fall within this threshold; red points fall outside of it.



Figure S4.4. Relative error (%) in speed & distance estimates across different sampling intervals for simulated Mongolian gazelles (*Procapra gutturosa*) for a leave-one-out approach. Points represent the mean relative error (%) for one of 50 iterations (excluding a different individual each time), with its 95% confidence intervals. Horizontal dotted lines indicate the $\pm 5\%$ error threshold. Blue points fall within this threshold; red points fall outside of it.